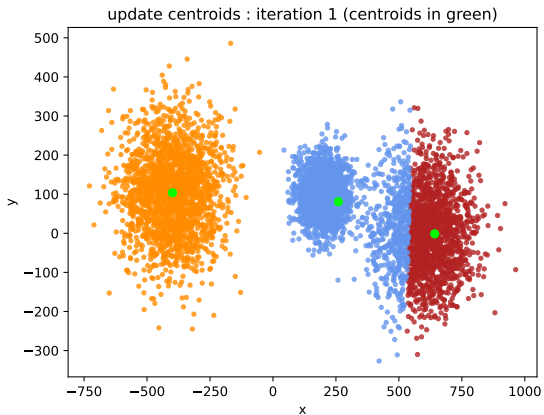


Machine learning Tek 5: clustering



Motivation

K-means clustering

Hierarchical clustering

Spectral clustering

Motivation

K-means clustering

Hierarchical clustering

Spectral clustering

Partitions

- ▶ **Example 1** : A is the set of even integers, B the set of odd integers. Is (A, B) a partition of $\mathbb{N}$?
- ▶ **Example 2** : C is the set of multiples of 2, D the set of multiples of 3. Is (C, D) a partition of $\mathbb{N}$?

Example : partition of data

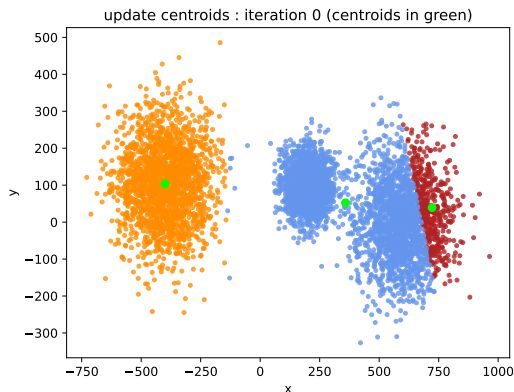


Figure – In this image, each cluster is represented by a color.

Applications of clustering

Example applications :

- ▶ spam filtering [Sharma and Rastogi, 2014,]
- ▶ fake news identification
[Hosseinimotlagh and Papalexakis, 2018,]
- ▶ marketing and sales
- ▶ document analysis [Zhao and Karypis, 2002,]
- ▶ traffic classification [Woo et al., 2007,]

Some of these applications can be considered to be semi-supervised learning.

Clustering is wide

- ▶ Applications :
 - ▶ https://en.wikipedia.org/wiki/Cluster_analysis
 - ▶ <https://datafloq.com/read/7-innovative-uses-of-clustering-algorithms/>
- ▶ Many clustering algorithms exist !
<https://scikit-learn.org/stable/modules/clustering.html>

K means clustering

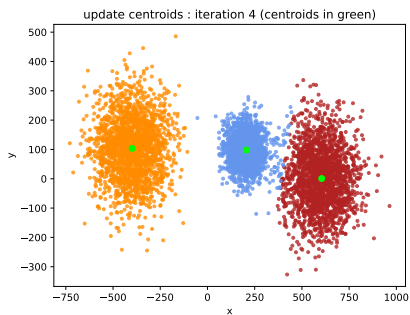


Figure – K means clustering

K-means

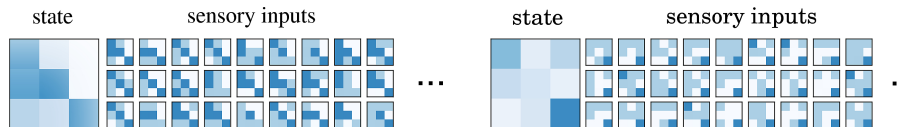


Figure – Other example of k-means clustering, this time in 9 dimensions
[Le Hir et al., 2018]

K-means : Expectation Maximisation algorithm

- ▶ Classical Machine Learning algorithm (EM)
- ▶ Discussion on the drawbacks of the algorithm.

K-means clustering

Exercise 1 : Implementing kmeans

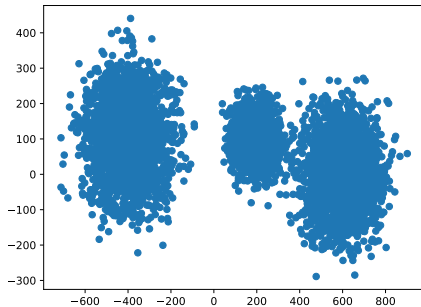


Figure – Data we want to cluster

K-means clustering

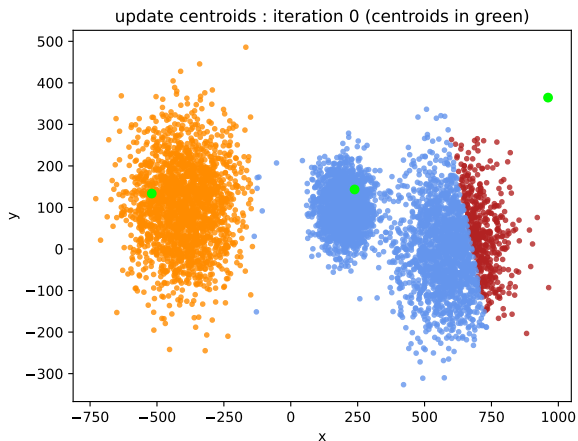
Exercise 1 : Implementing k-means

`cd 4_clustering/1_k_means/`.

Edit the `k_means.py` file so that it performs the k-means algorithm, on the example dataset, with 3 clusters.

You can use for loops or try to directly write the algorithm in a vectorized way.

K-means



K-means

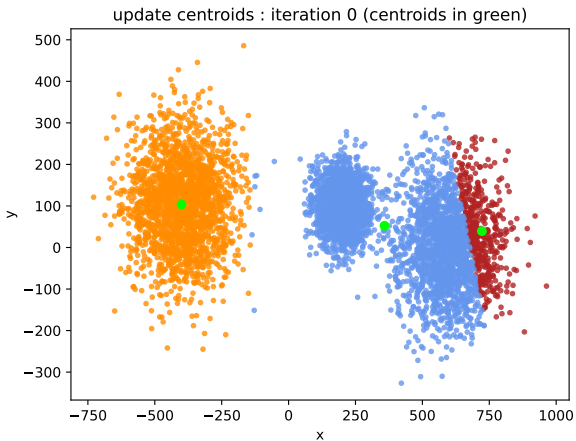
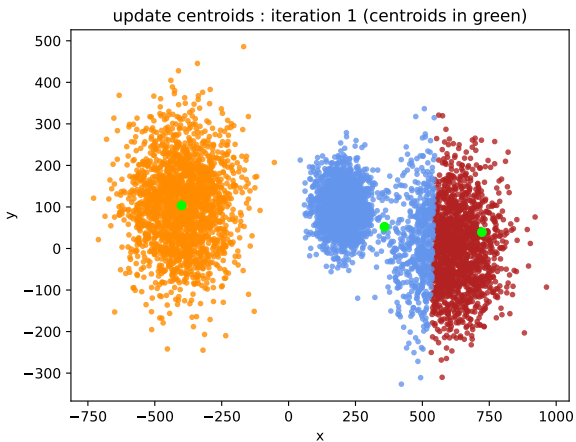


Figure – Centroids 0th iteration

K-means



Figure

K-means

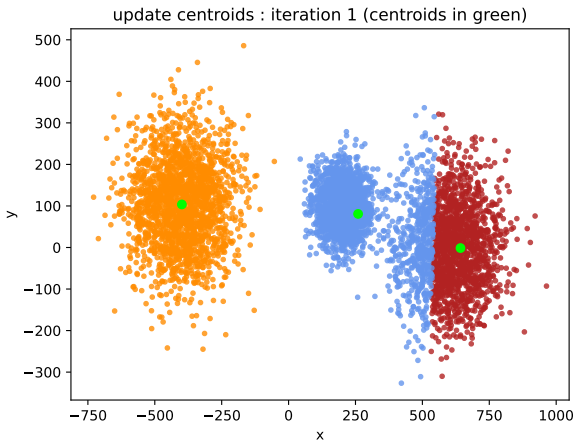
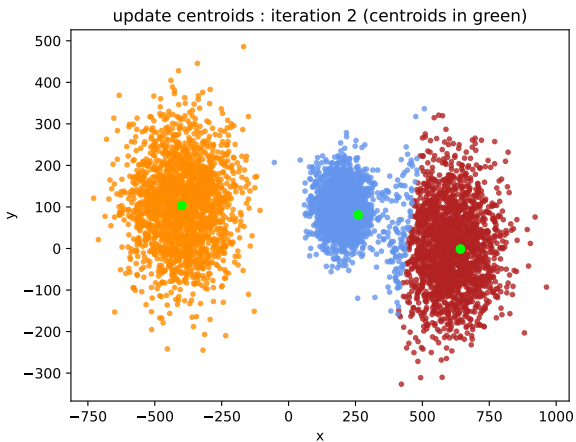


Figure – Centroids 1st iteration

K-means



Figure

K-means

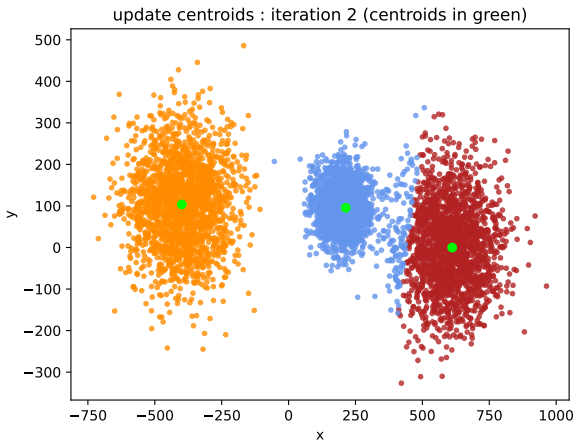
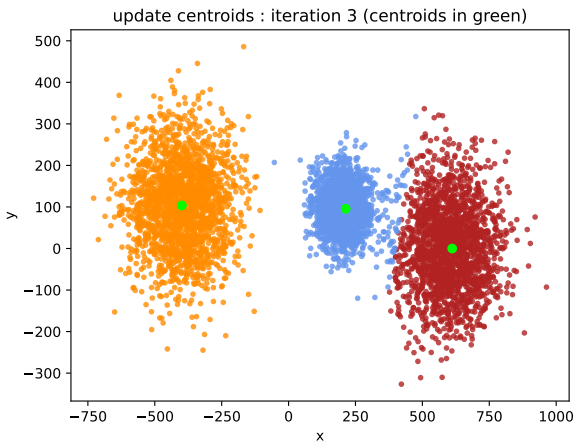


Figure – Centroids 1st iteration

K-means



Figure

K-means

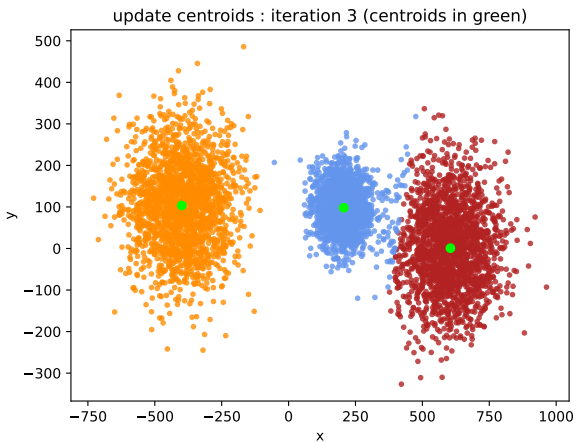


Figure – Centroids 1st iteration

K-means and initialization

Note that when launching the algorithm several times, the result may differ. **Why ?**

K-means optimization problem

- ▶ When performing the k-means algorithm, we optimize the **inertia**.
- ▶ if we have n points, x_i , each assigned to a centroid c_i .

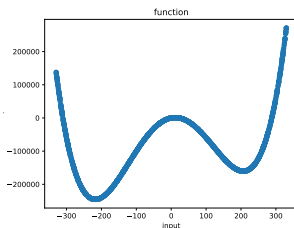
$$I = \sum_{i=1}^n d(x_i, c_i)^2 \quad (4)$$

$$I = \sum_{i=1}^n \|x_i - c_i\|_2^2 \quad (5)$$

$\|$ stands for "norm".

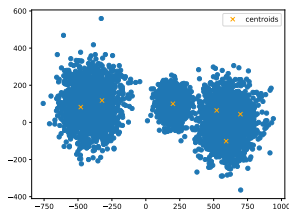
K-means : Expectation Maximisation algorithm

- What could we do if the algorithm falls in a local optimum?



Exercise 2: Perform the algorithm on the same dataset with scikit-learn (use the file : `k_means_sklearn.py`)

- ▶ Observe the randomness of the result
- ▶ Explore the available parameters :
<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html> and find a solution in order to observe a **stable** result.



Knee/elbow criterion

- ▶ We would like a **heuristic method** in order to be able to assess a relevant number of clusters.

Knee/elbow criterion

- ▶ **Exercise 3:** use the file `k_means_inertia.py` in order to find a relevant number of clusters for the `data_3D.npy`, with scikit-learn and kneed.

<https://github.com/arvkevi/kneed>

Observe the behavior of this method when you run it on a different input dataset (you can generate a new one).

[https://scikit-learn.org/stable/auto_examples/cluster/
plot_kmeans_assumptions.html](https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_assumptions.html)

Motivation

K-means clustering

Hierarchical clustering

Spectral clustering

Hierarchies

It is possible to perform a clustering in a hierarchical way. This means building a **sequence of clusterings**.

https://scikit-learn.org/stable/auto_examples/cluster/plot_agglomerative_dendrogram.html

<https://docs.scipy.org/doc/scipy/reference/cluster.hierarchy.html>

Example application of hierarchical clustering

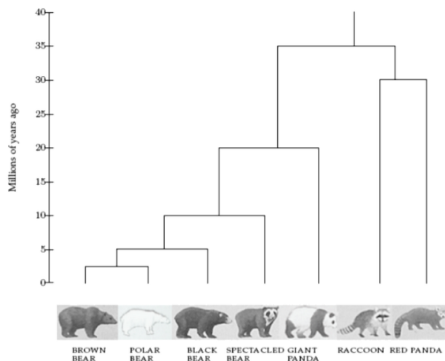


Figure – <https://towardsdatascience.com/hierarchical-clustering-and-its-applications-41c1ad4441a6>

Hierarchical clustering

We will apply hierarchical clustering to a small example dataset containing addresses.

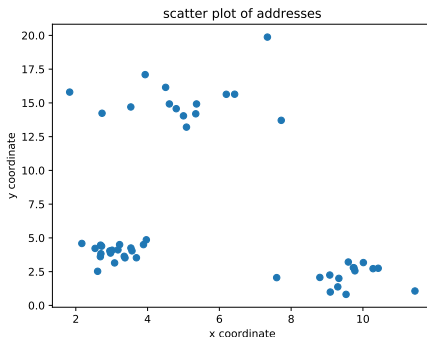
Hierarchical clustering

Exercise 4 : Plotting data

`cd hierarchical_clustering/` and use `hierarchical_clustering.py` in order to show the scatter plot of the data (nuage de points) loaded from `addresses.csv`.

Seaborn lib : <https://seaborn.pydata.org/>

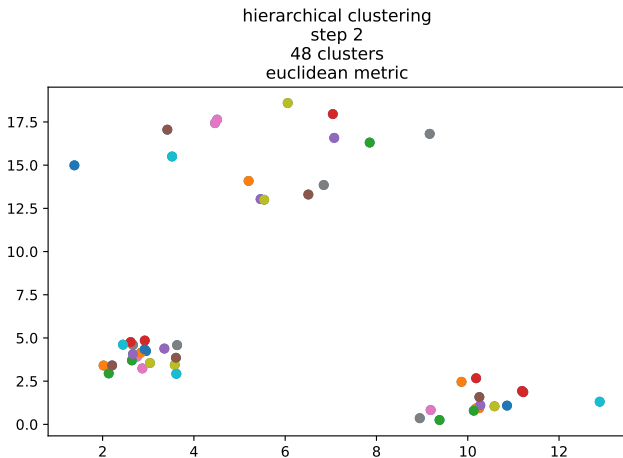
Hierarchical clustering

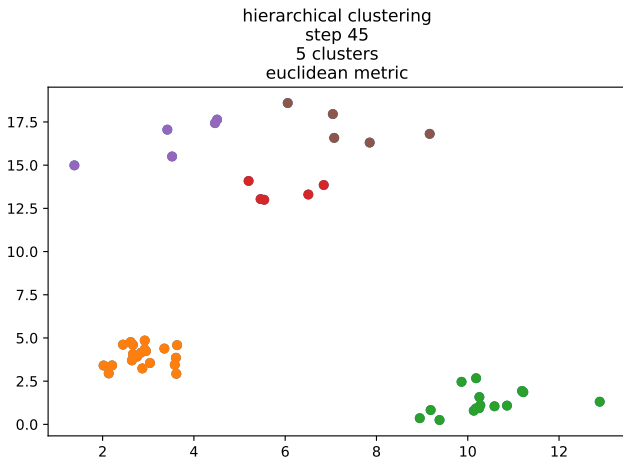


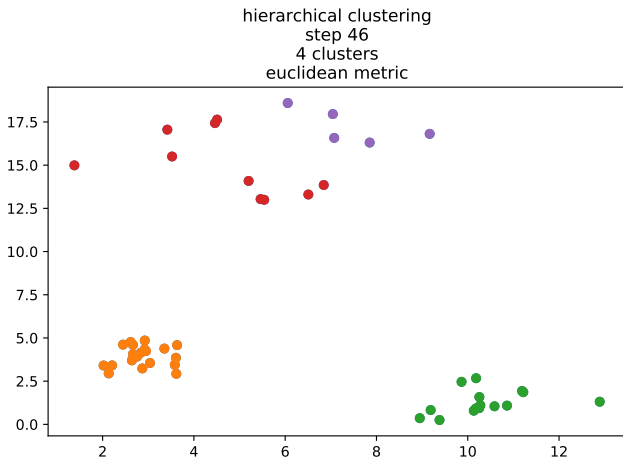
Hierarchical clustering consists in progressively grouping points together in **classes**.

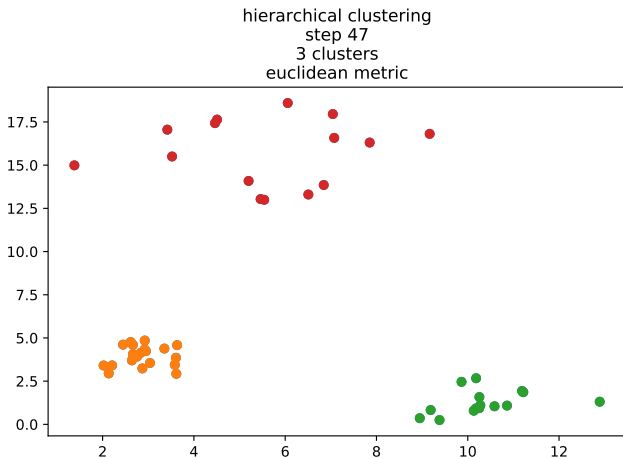
Hierarchical clustering

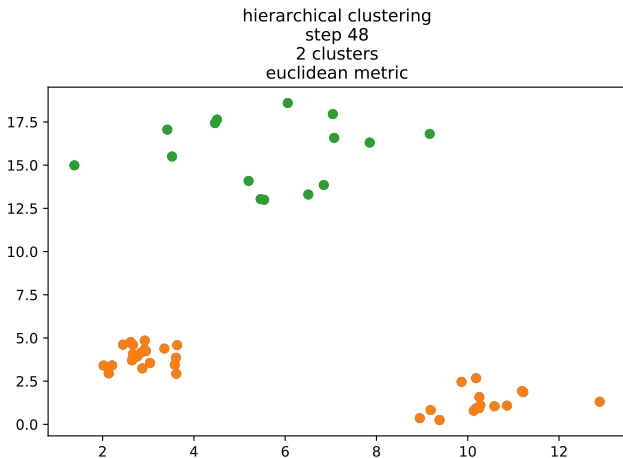
Exercise 5: Edit `find_closest_classes()` in order to find the closest classes. Then they can be merged in the while loop.

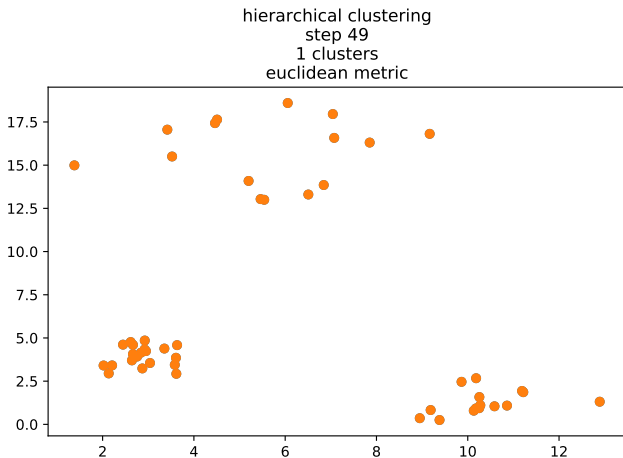












Hierarchical clustering

An important aspect of hierarchical clustering is that different criteria can be used in order to merge the classes. The distance between class 1 and class 2 can for instance be :

- ▶ the minimum distance between one point of class 1 and one point of class 2 : **single-linkage clustering**.
- ▶ the average distance between points a class 1 and points a class 2 : **unweighted average linkage clustering**

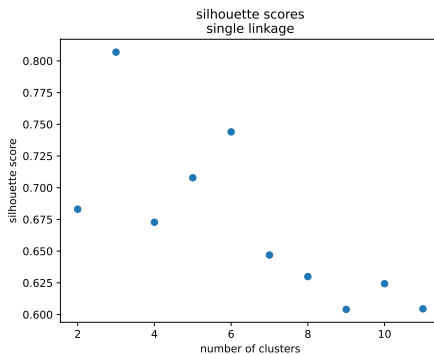
The two methods can lead to a different hierarchy of clusters.

Average-linkage clustering

Exercice 6 : Modify the computation of the distance between classes using the average-linkage criterion.

Silhouette score

The Silhouette score is another heuristic used in order to choose a relevant number of clusters. File : **hierarchical_clustering_silhouette.py**.



Motivation

K-means clustering

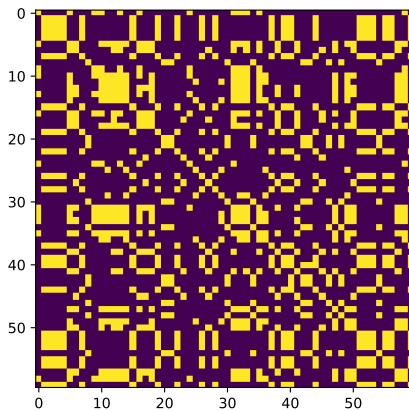
Hierarchical clustering

Spectral clustering

Similarities

- ▶ When working with distances, two points that "look the same" should be separated by a **small distance**.
- ▶ When working with a similarity, two points that "look the same" should have a **high similarity**.
- ▶ Example : **adjacency relationship** in a graph : if i and j are related by an edge, $S_{ij} = 1$, otherwise $S_{ij} = 0$.

Adjacency matrix



Similarities

Differences between similarities and distances :

- ▶ A similarity S is not always symmetrical.
- ▶ Indeed, in a **directed graph**, having a directed edge between i and j does not mean that we have an edge between j and i .
- ▶ $S_{ij} = 0$ does not mean that $i = j$, it is rather the contrary.

Similarities

- ▶ A similarity is a more general notion than a distance. Given a similarity between two points, we can deduce a distance.
- ▶ For instance this way, if d_{ij} is the distance between i and j :

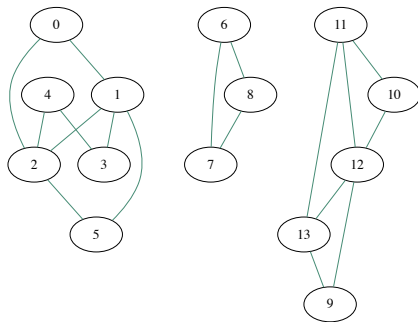
$$S_{ij} = \exp(-d_{ij}) \quad (6)$$

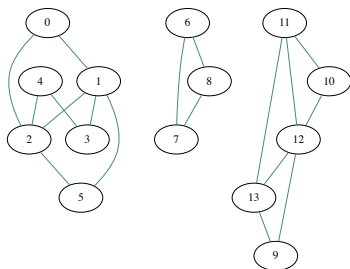
Spectral Clustering

- ▶ A clustering method that works with similarities
- ▶ It performs a low dimensional embedding of the similarity matrix, followed by a Kmeans

Exercise

We will perform Spectral Clustering on this graph :





cd `spectral_clustering/` and use `vanilla_spectral_clustering.py` in order to apply spectral clustering. You first need to input the right **affinity matrix** or **similarity matrix** and then use the **scikit-learn** library. **doc** : check the scikit page for Spectral Clustering.

Spectral clustering

Some drawbacks of the method :

- ▶ Need to provide the number of clusters.
- ▶ Not adapted to a large number of clusters.
- ▶ kmeans step : so depends on a random initialization.

Heuristic

- ▶ We would like a criterion in order to justify the number of clusters used.

Normalized cut : a measurement of the quality of a clustering

- ▶ The **cut of a cluster** C_k , noted $Cut(C_k, V \setminus C_k)$ is the number of outside connections (connections with other clusters).
- ▶ The **degree** of a node is its number of adjacent edges
- ▶ The **degree of a cluster**, noted d_{C_k} is the sum of the degrees of its nodes.
- ▶ The **normalized cut** of a clustering that contains K clusters is :

$$NCut(C) = \sum_{k=1}^K \frac{Cut(C_k, V \setminus C_k)}{d_{C_k}} \quad (7)$$

Normalization

- ▶ The normalization is useful in order to take the **weight** (degree) of a cluster into account.

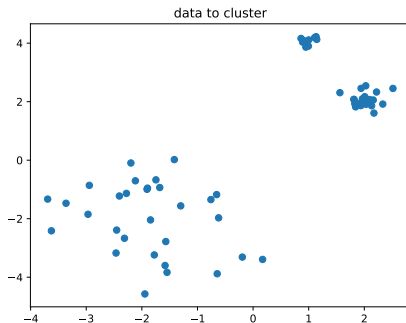
Normalized cut and clustering

Let's see how the normalized cut can help us choose the right number of clusters (backboard).

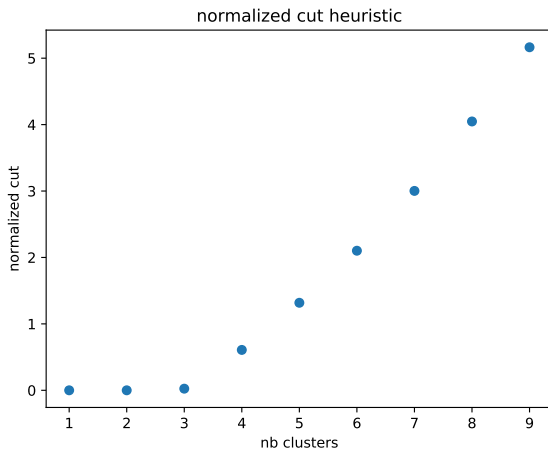
Heuristic

Exercise 7 : Normalized but elbow :

Use `normalized_cut.py` in order to estimate the relevant number of clusters in order to process the data contained in `data/`. These data are generated by `generate_data.py`.



Normalized cuts



Normalized cuts

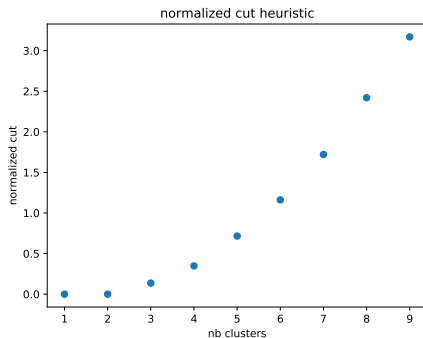
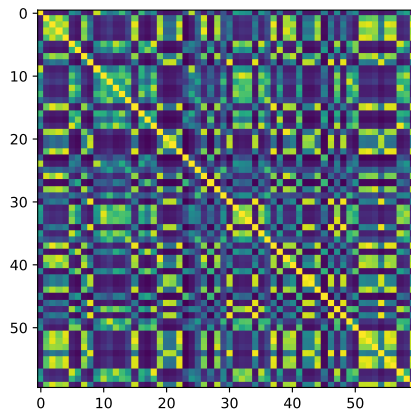


Figure – If the standard deviations in the dataset are larger, it is harder to identify a relevant number of clusters.

Similarity



Example

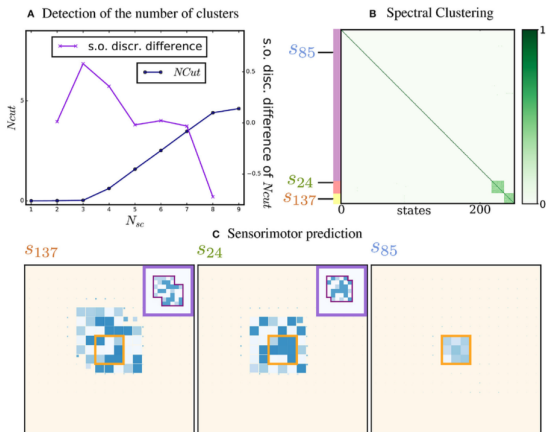


Figure – In a), the elbow method is used to choose the number of clusters. [Le Hir et al., 2018]

Other methods to evaluate the quality of a clustering

- ▶ Stability of the result when launching the algorithm many times
- ▶ Separation of the clusters (the mean distance between pairs of centroids is large)
- ▶ Ratio inter / intra
- ▶ Silhouette coefficient

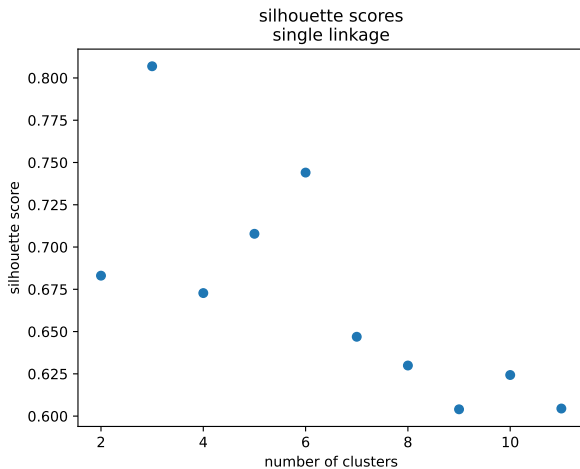
`https://scikit-learn.org/stable/modules/clustering.html`

Exercise 8 :

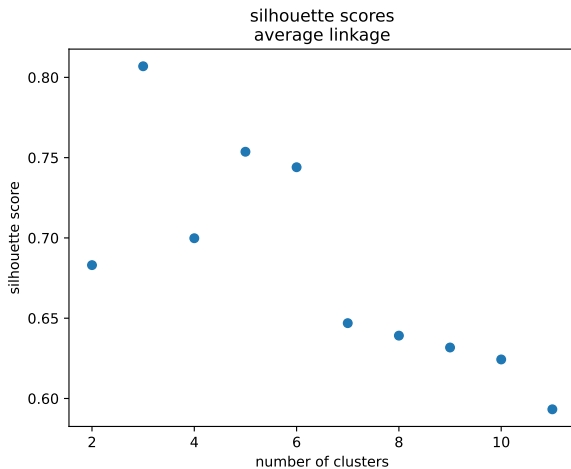
Study the silhouette score as a function of the number of clusters for the hierarchical clustering problem.

`https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette\_score.html`

Silhouette coefficient with single linkage



Silhouette coefficient with average linkage



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